

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. MINOR IN INTERNET OF THINGS**

**COURSE STRUCTURE AND SYLLABUS**

(Applicable for the batches admitted from the academic year 2022-2023)

**V SEMESTER**

**R22**

| Course Code  | Title of the Course              | L        | T        | P/D      | CH       | C          |
|--------------|----------------------------------|----------|----------|----------|----------|------------|
| 22MC1IN301   | Programming in Python            | 3        | 0        | 0        | 3        | 3          |
| 22MC2IN301   | Programming in Python Laboratory | 0        | 0        | 3        | 3        | 1.5        |
| <b>Total</b> |                                  | <b>3</b> | <b>0</b> | <b>3</b> | <b>6</b> | <b>4.5</b> |

**VI SEMESTER**

**R22**

| Course Code  | Title of the Course               | L        | T        | P/D      | CH       | C        |
|--------------|-----------------------------------|----------|----------|----------|----------|----------|
| 22MC1IN302   | IoT Fundamentals and Architecture | 3        | 0        | 0        | 3        | 3        |
| 22MC1IN303   | Smart Technologies                | 3        | 0        | 0        | 3        | 3        |
| <b>Total</b> |                                   | <b>6</b> | <b>0</b> | <b>0</b> | <b>6</b> | <b>6</b> |

**VII SEMESTER**

**R22**

| Course Code  | Title of the Course                            | L        | T        | P/D      | CH       | C          |
|--------------|------------------------------------------------|----------|----------|----------|----------|------------|
| 22MC1IN401   | Programming Languages for IoT                  | 3        | 0        | 0        | 3        | 3          |
| 22MC2IN401   | IoT Automation using Raspberry – PI Laboratory | 0        | 0        | 3        | 3        | 1.5        |
| <b>Total</b> |                                                | <b>3</b> | <b>0</b> | <b>3</b> | <b>6</b> | <b>4.5</b> |

**VII SEMESTER**

**R22**

| Course Code  | Title of the Course            | L        | T        | P/D      | CH       | C        |
|--------------|--------------------------------|----------|----------|----------|----------|----------|
| 22MC1IN403   | Fog and Edge Computing for IoT | 3        | 0        | 0        | 3        | 3        |
| <b>Total</b> |                                | <b>3</b> | <b>0</b> | <b>0</b> | <b>3</b> | <b>3</b> |

L – Lecture

T – Tutorial

P – Practical

D – Drawing

CH – Contact Hours/Week

C – Credits

SE – Sessional Examination

CA – Class Assessment

ELA – Experiential Learning Assessment

SEE – Semester End Examination

D-D – Day to Day Evaluation

LR – Lab Record

CP – Course Project

PE – Practical Examination

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) V Semester

#### (22MC1IN301) PROGRAMMING IN PYTHON

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To learn syntax and semantics and create functions in Python
- To handle strings and files in Python
- To understand lists, dictionaries and regular expressions in Python
- To implement Object Oriented Programming concepts in Python
- To gain knowledge on Web Services, network programming and Database Programming in Python

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Examine Python syntax and semantics and be fluent in the use of Python flow control and functions

**CO-2:** Demonstrate proficiency in handling Strings and File Systems

**CO-3:** Create, run, and manipulate Python Programs using core data structures like Lists, Dictionaries and use Regular Expressions

**CO-4:** Interpret the concepts of Object-Oriented Programming as used in Python

**CO-5:** Implement exemplary applications related to Network Programming, Web Services and Databases in Python

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 |
| CO-1 | 1                     | 2    | 2    | 2    | 3    | 2    | 2    | -    | 1    | -     | -     | 1     |
| CO-2 | 3                     | 3    | 2    | 2    | 3    | 1    | 2    | -    | 2    | -     | -     | 2     |
| CO-3 | 2                     | 2    | 3    | 3    | 3    | 2    | 2    | -    | 2    | -     | --    | 2     |
| CO-4 | 2                     | 3    | 3    | 3    | 3    | 1    | 3    | -    | 2    | -     | -     | 2     |
| CO-5 | 3                     | 3    | 3    | 3    | 3    | 2    | 2    | 1    | 3    | 3     | 3     | 3     |

#### UNIT-I:

**Python:** Basics, Objects- Python Objects, Standard Types, Other Built-in Types, Internal Types, Standard Type Operators, Standard Type Built-in Functions, Categorizing the Standard Types, Unsupported Types

**Numbers** - Introduction to Numbers, Integers, Floating Point Real Numbers, Complex Numbers, Operators, Built-in Functions, Related Modules

Sequences - Strings, Lists, and Tuples, Mapping and Set Types

## **UNIT-II:**

**Files:** File Objects, File Built-in Function [ open() ], File Built-in Methods, File Built-in Attributes, Standard Files, Command line Arguments, File System, File Execution, Persistent Storage Modules, Related Modules

**Exceptions:** Exceptions in Python, Detecting and Handling Exceptions, Context Management, \*Exceptions as Strings, Raising Exceptions, Assertions, Standard Exceptions, \*Creating Exceptions, Why Exceptions (Now)?, Why Exceptions at All?, Exceptions and the sys Module, Related Modules

**Modules:** Modules and Files, Namespaces, Importing Modules, Importing Module Attributes, Module Built-in Functions, Packages, Other Features of Modules.

## **UNIT-III:**

**Regular Expressions:** Introduction, Special Symbols and Characters, Res and Python

**Multithreaded Programming:** Introduction, Threads and Processes, Python, Threads, and the Global Interpreter Lock, Thread Module, Threading Module, Related Modules

## **UNIT-IV:**

**GUI Programming:** Introduction, Tkinter programming, Tkinter widgets, Frame, Button, Label, Entry, Brief Tour of Other GUIs.

**WEB Programming:** Introduction, Architecture, CGI environment variable, GET and POST methods, Cookies, File upload, Web Surfing with Python, Creating Simple Web Clients, Advanced Web Clients, CGI-Helping Servers Process Client Data, Building CGI Application, Advanced CGI, Web (HTTP) Servers

## **UNIT-V:**

**Database Programming:** Introduction, Python Database Application Programmer's Interface (DB-API), Object Relational Managers (ORMs), Related Modules.

## **TEXT BOOKS:**

1. Core Python Programming, Wesley J. Chun, 2<sup>nd</sup> Edition, Pearson
2. Data Structures and Algorithm Analysis in C, Mark Allen Weiss, 2<sup>nd</sup> Edition, Pearson Education, 2010
3. Think Python: How to Think Like a Computer Scientist, Allen B. Downey, 2<sup>nd</sup> Edition, Updated for Python 3, Shroff/O'Reilly, 2016 (<http://greenteapress.com/wp/thinkpython/>)

## **REFERENCES:**

1. Data Structures and Algorithms, A. V. Aho, J. E. Hopcroft, and J. D. Ullman, Pearson Education, 1<sup>st</sup> Edition Reprint 2003
2. Introduction to Computer Science using Python: A Computational Problem-Solving Focus, Charles Dierbach, Wiley India Edition, 2013
3. Practical Programming: An Introduction to Computer Science using Python 3, Paul Gries, Jennifer Campbell and Jason Montojo, 2<sup>nd</sup> Edition, Pragmatic Programmers, LLC, 2013

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) V Semester

#### (22MC2IN301) PROGRAMMING IN PYTHON LABORATORY

| TEACHING SCHEME |     |     |
|-----------------|-----|-----|
| L               | T/P | C   |
| 0               | 3   | 1.5 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand program design with functions using Python programming language
- To understand a range of Object-Oriented Programming, as well as in-depth data and information processing techniques
- To understand Web Scraping using Python
- To understand the high-performance programs designed to strengthen the practical expertise
- To gain knowledge on Turtle graphics library

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Develop the python scripts using regular expressions

**CO-2:** Explore python especially the object-oriented concepts, and the built-in objects of Python

**CO-3:** Develop GUI Application using Python tkinter

**CO-4:** Create practical and contemporary applications such as Web applications, discrete event simulations

**CO-5:** Implement python program using the Turtle graphics library

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 |
| CO-1 | 1                     | 2    | 3    | 2    | 2    | 2    | 2    | -    | 1    | -     | -     | 1     |
| CO-2 | 3                     | 3    | 3    | 3    | 2    | 1    | 2    | -    | 2    | -     | -     | 2     |
| CO-3 | 3                     | 2    | 3    | 2    | 2    | 2    | 2    | -    | 2    | -     | -     | 2     |
| CO-4 | 3                     | 2    | 3    | 3    | 3    | 1    | 2    | -    | 2    | -     | -     | 2     |
| CO-5 | 3                     | 3    | 3    | 3    | 2    | 2    | 2    | 1    | 2    | 3     | 3     | 3     |

#### LIST OF PROGRAMS:

1. Python program to calculate student grade.
2. Python example to implement destructor and constructors using `__del__()` and `__init__()`
3. Python example to implement Getters and Setters in a class
4. Python - Implement Abstraction using Abstract class
5. Python - Implement Interface using class

6. Python - Create Employee Class with Constructor and Destructor
7. Example of single inheritance in Python
8. Python program to illustrate Single Inheritance
9. Example of multiple inheritance in Python
10. Example of Multilevel Inheritance in Python
11. Example of Hierarchical Inheritance in Python
12. Using Regular Expressions, develop a Python program to
  - a) Identify a word with a sequence of one upper case letter followed by lower case letters.
  - b) Find all the patterns of "1(0+)1" in a given string.
  - c) Match a word containing 'z' followed by one or more o's. Prompt the user for input.
13. GUI&DB
  - a) Design a GUI based calculator to perform arithmetic operations like addition, subtraction, multiplication and division. (Hint: Expression Calculator using tk)
  - b) Design a GUI based application to convert temperature from Celsius to Fahrenheit.
  - c) Write a python program to perform various database operations (create, insert, delete, update).
14. GRAPHICS
  - a) Consider turtle object. Write functions to draw triangle, rectangle, polygon, circle and sphere. Use object-oriented approach.
  - b) Design a Python program using the Turtle graphics library to construct a turtle bar chart representing the grades obtained by N students read from a file categorizing them into distinction, first class, second class, third class and failed.

#### **TEXT BOOKS:**

1. Core Python Programming, Wesley J. Chun, 2<sup>nd</sup> Edition, Pearson
2. Data Structures and Algorithm Analysis in C, Mark Allen Weiss, 2<sup>nd</sup> Edition, Pearson Education, 2010
3. Think Python: How to Think Like a Computer Scientist, Allen B. Downey, 2<sup>nd</sup> Edition, Updated for Python 3, Shroff/O'Reilly, 2016 (<http://greenteapress.com/wp/thinkpython/>)

#### **REFERENCES:**

1. Data Structures and Algorithms, A. V. Aho, J. E. Hopcroft, and J. D. Ullman, Pearson Education, 1<sup>st</sup> Edition Reprint 2003
2. Introduction to Computer Science using Python: A Computational Problem-Solving Focus, Charles Dierbach, Wiley India, 2013
3. Practical Programming: An Introduction to Computer Science using Python 3, Paul Gries, Jennifer Campbell and Jason Montoyo, 3<sup>rd</sup> Edition, Pragmatic Programmers, 2017

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) VI Semester

#### (22MC1IN302) IOT FUNDAMENTALS AND ARCHITECTURE

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce terminology, technology and its applications
- To learn Raspberry PI platform, that is widely used in IoT applications
- To understand the implementation of web-based services on IoT devices
- To gain knowledge on the internet of things and its characteristics
- To learn concepts on API web servers

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Understand IoT value chain structure (device, data cloud), application areas and technologies involved

**CO-2:** Understand IoT sensors and technological challenges faced by IoT devices, with a focus on wireless, energy, power, and sensing modules

**CO-3:** Implement basic IoT applications on embedded platform

**CO-4:** Understand building blocks of Internet of Things and characteristics

**CO-5:** Analyze the concepts on API web servers.

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 |
| CO-1 | 1                     | 1    | 3    | 2    | 3    | 2    | 2    | -    | 1    | -     | -     | 1     |
| CO-2 | 3                     | 3    | 3    | 2    | 2    | 1    | 2    | -    | 2    | -     | -     | 2     |
| CO-3 | 3                     | 2    | 3    | 3    | 2    | 2    | 2    | -    | 2    | -     | -     | 2     |
| CO-4 | 3                     | 2    | 3    | 2    | 3    | 1    | 3    | -    | 2    | -     | -     | 2     |
| CO-5 | 3                     | 3    | 3    | 3    | 3    | 2    | 2    | 1    | 3    | 3     | 3     | 3     |

#### UNIT-I:

**Introduction to Internet of Things:** Definition and Characteristics of IoT, Sensors, Actuators, Physical Design of IoT – IoT Protocols, IoT communication models, IoT Communication APIs, IoT enabled Technologies – Wireless Sensor Networks, Cloud Computing, Embedded Systems, IoT Levels and Templates, Domain Specific IoTs – Home, City, Environment, Energy, Agriculture and Industry.

## **UNIT-II:**

**IoT and M2M:** Software defined networks, network function virtualization, difference between SDN and NFV for IoT, Basics of IoT System Management with NETCOZF, YANG-NETCONF, YANG, SNMP NETOPEER

## **UNIT-III:**

**IoT Physical Devices and Endpoints:** Introduction to Arduino and Raspberry Pi- Installation, Interfaces (serial, SPI, I2C), Programming – Python program with Raspberry PI with focus on interfacing external gadgets, controlling output, reading input from pins.

## **UNIT-IV:**

**Controlling Hardware:** Connecting LED, Buzzer, Switching High Power devices with transistors, Controlling AC Power devices with Relays, Controlling servo motor, speed control of DC Motor, unipolar and bipolar Stepper motors.

**Sensors:** Light sensor, temperature sensor with thermistor, voltage sensor, ADC and DAC, Temperature and Humidity Sensor DHT11, Motion Detection Sensors, Wireless Bluetooth Sensors, Level Sensors, USB Sensors, Embedded Sensors, Distance Measurement with ultrasound sensor

## **UNIT-V:**

**IoT Physical Servers and Cloud Offerings:** Introduction to Cloud Storage models and communication APIs Webserver – Web server for IoT, Cloud for IoT, Python web application framework Designing a RESTful web API

## **TEXT BOOKS:**

1. Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015
2. Getting Started with Raspberry Pi, Matt Richardson & Shawn Wallace, O'Reilly, 2014
3. Raspberry Pi Cookbook, Software and Hardware Problems and solutions, Simon Monk, O'Reilly, 2016

## **REFERENCES:**

1. Learning Internet of Things, Peter Waher, Ovidiu Vermesan, Packt Publishing, 2015
2. Internet of Things – From Research and Innovation to Market Deployment, Peter Friess, River Publishers, 2014
3. Actuators and Their Interfaces, N. Ida, Sensors, SciTech Publishers, 2014

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) VI Semester

#### (22MC1IN303) SMART TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To provide a detailed description of the integral aspects of smart technologies and their evolution to their current state
- To discuss the potential use of Internet of things (IoT) in reducing counterproductive work behaviours and identifying some of the challenges that organizations might face while implementing IoT in its systems
- To present case studies using easy-to-understand language to explain the breadth and scope of application areas
- To learn the knowledge on cyber care
- To understand the importance of NICE labs

**COURSE OUTCOMES:** After completion of course, the student should be able to

**CO-1:** Explain importance, scope and applications of Smart Technologies

**CO-2:** Implement Drones an IoT based services and block chain technologies

**CO-3:** Provide solutions for real world IoT applications like vehicle cyber security

**CO-4:** Understand the importance of cyber care and develop IoT-Based Smart Farming Solutions

**CO-5:** Analyze case study on NICE Labs

#### COURSE ARTICULATION MATRIX:

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)*

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 |
| CO-1 | 1                     | 1    | 3    | 2    | 3    | 2    | 2    | -    | 1    | -     | -     | 1     |
| CO-2 | 2                     | 3    | 3    | 2    | 2    | 1    | 2    | -    | 2    | -     | -     | 2     |
| CO-3 | 2                     | 2    | 3    | 3    | 2    | 2    | 2    | -    | 2    | -     | -     | 2     |
| CO-4 | 2                     | 2    | 3    | 2    | 3    | 1    | 3    | -    | 2    | -     | -     | 2     |
| CO-5 | 3                     | 3    | 3    | 3    | 3    | 3    | 3    | 1    | 2    | 3     | 3     | 2     |

#### UNIT-I:

**Smart Technologies:** Scope and Applications, Cutting-Edge Digitization Challenges in Vehicle Cyber Physical Systems and Cybersecurity, Big Data Analytics as an Enabler in Smart Governance for the Future Smart Cities



**UNIT-II:**

**Digital Masters:** Blueprinting Digital Transformation, UAVs/Drones-Based IoT Services, Role of Cyber Security in Drone Technology

Bitcoins as an Implementation of Blockchain and Its Convergence with Internet of Things, Tomorrow's AI-Enabled Banking, Exploring Connected Cars

**UNIT-III:**

Vehicular Cybersecurity Through Intrusion Detection and Prevention Architecture, Mechanism Protecting Vehicle-to-Vehicle Communication, Advanced Driver Assistance Systems

**UNIT-IV:**

**Cyber Care:** Role of Cyber Security in Healthcare Industry, Smart Agriculture: A Tango Between Modern IoT-Based Technologies and Traditional Agriculture Techniques.

**UNIT-V:**

Importance of Being 'NICE' While Developing IoT-Based Smart Farming Solutions: A Case Study About 'NICE' Labs.

**TEXT BOOK:**

1. Smart Technologies-Scope and Applications, K. B. Akhilesh, Dietmar P. F. Möller, Springer, 2020

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) VII Semester

#### (22MC1IN401) PROGRAMMING LANGUAGES FOR IOT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To use more computing intensive IoT applications as examples using Raspberry Pi running Linux as the platform of choice
- To learn basic programming instructions on Raspberry Pi
- To understand IoT design using Raspberry Pi

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Appreciate the development technology for IoT

**CO-2:** Familiar with Basic Concepts of Linux

**CO-3:** Design real time IoT Devices

**CO-4:** Familiar with basic foundations of Python Programming and libraries

**CO-5:** Comprehend the basic concepts of Mobile Cloud Computing

#### UNIT-I:

**Getting Started with Raspberry Pi:** Basic functionality of Raspberry Pi B+ board, setting up the board, configuration and use, implications of an operating system on the behaviour of the Raspberry Pi as an IoT device, booting Raspberry Pi 3, Downloading an Operating System, format an SD card and booting the OS, Basics of Linux and its use, main features including navigating the file system and managing processes, text based user interface through the shell, overview of the graphic user interface for Raspian Linux distribution.

#### UNIT-II:

Interfacing Hardware with the Raspberry Pi, Raspberry Pi Remote Access, operates the Raspberry Pi in "headless mode", Bash Command line, operating Raspberry Pi without needing a GUI interface.

Basics of the Python programming language, programming on the Raspberry Pi. Python on Raspberry Pi, Python Programming Environment, Python Expressions, Strings, Functions and Function arguments, Lists, List Methods, Control Flow.

#### UNIT-III:

Communication with devices through the pins of the Raspberry Pi, RPi. GPIO library, Python Functions, setting up the pins, General purpose IO Pins, Protocol Pins, GPIO Access, applying digital voltages, and generating Pulse Width Modulated signals, Tkinter Python library, accessing pins through a graphic user interface

#### UNIT-IV:

**IoT Physical Servers and Cloud Offerings:** Introduction to Cloud Storage models and communication APIs. Web Server – Web server for IoT, Cloud for IoT, Python web application framework. Designing a RESTful web API. Connecting to APIs

**UNIT-V:**

IoT Design using Raspberry Pi IoT Applications based on Pi, LAMP Web-server, GPIO Control over Web Browser, Creating Custom Web Page for LAMP, communicating data using on-board module, home automation using Pi, Node-RED, MQTT Protocol, Using Node-RED Visual Editor on Rpi

**TEXT BOOKS:**

1. Programming the Raspberry Pi: Getting Started with Python, Simon Monk, McGraw-Hill Professional, 2012
2. Raspberry Pi with Java: Programming the Internet of Things, 1<sup>st</sup> Edition, Oracle Press
3. The Official Raspberry Pi Projects Book, [https://www.raspberrypi.org/magpiissues/Projects\\_Book\\_v1.pdf](https://www.raspberrypi.org/magpiissues/Projects_Book_v1.pdf)

**REFERENCE BOOKS:**

1. Raspberry Pi User Guide, Eben Upton and Gareth Halfacree, 4<sup>th</sup> Edition, John Wiley, 2016
2. Learning Python with Raspberry Pi, Alex Bradbury and Ben Everard, John Wiley, 2014
3. Arduino Cookbook, Michael Margolis, 1<sup>st</sup> Edition, O'Reilly, 2011

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) VII Semester

#### (22MC2IN401) IOT AUTOMATION USING RASPBERRY-PI LABORATORY

| TEACHING SCHEME |     |     |
|-----------------|-----|-----|
| L               | T/P | C   |
| 0               | 3   | 1.5 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

#### COURSE OBJECTIVES:

- To introduce the raspberry PI platform, that is widely used in IoT applications
- To introduce the implementation of distance sensor on IoT devices
- To understand installation of OS in Raspberry PI
- To use raspberry PI for implementing alarmed motion sensor
- To provide hands-on experience in integrating Raspberry Pi with wired and wireless communication protocols

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Introduce the concept of M2M (machine to machine) with necessary protocols and get awareness in implementation of distance sensor

**CO-2:** Program using python scripting language which is used in many IoT devices

**CO-3:** Implement alarmed motion sensor using Raspberry PI

**CO-4:** Install Operating System in Raspberry PI

**CO-5:** Design and develop real-time IoT automation systems using Raspberry Pi, sensors, and actuators

#### LIST OF EXPERIMENTS:

##### USING RASPBERRY PI

1. Calculate the distance using a distance sensor
2. Basic LED functionality
3. Calculate temperature using a temperature sensor
4. Build an alarmed motion sensor
5. Make printer wireless
6. Add a power button to Raspberry pi
7. Build a network game server
8. Make music with sony Pi
9. Interface Pi Camera module with Raspberry Pi
10. Installing OS on Raspberry Pi
  - a) Installation using Pilmager
  - b) Installation using image File

#### TEXT BOOKS:

1. Programming the Raspberry Pi: Getting Started with Python, Simon Monk, McGraw-Hill Professional, 2012
2. Raspberry Pi with Java: Programming the Internet of Things, 1<sup>st</sup> Edition, Oracle Press
3. The Official Raspberry Pi Projects Book, [https://www.raspberrypi.org/magpiissues/Projects\\_Book\\_v1.pdf](https://www.raspberrypi.org/magpiissues/Projects_Book_v1.pdf)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. Minor (IOT) VIII Semester

#### (22MC1IN402) FOG AND EDGE COMPUTING FOR IOT

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand new computing paradigms and challenges in federated edge resources
- To learn more about integrating IoT, Fog and cloud infrastructures, and optimization problems in Fog computing
- To introduce middleware for Edge Cloud architectures
- To provide a detailed explanation of Fog Computing realization for Big Data analytics
- To provide a detailed explanation of Smart Surveillance Video Stream Processing for tracking human objects

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the use of the IoT architecture with its entities and protocols, from the IoT devices

**CO-2:** Familiar with the basic concepts of IoT, Fog, Cloud Infrastructures, and its components and working of components and its performance

**CO-3:** Understand Optimization Problems in Fog and Edge Computing

**CO-4:** Explore Data Management, Big data analytics, and video processing concepts in Fog Computing

**CO-5:** Understand the role of Fog Computing in big data analytics and real-time applications like smart surveillance

#### UNIT-I:

**Internet of Things (IoT) and New Computing Paradigms:** Introduction, Relevant Technologies, Fog and Edge Computing Completing the Cloud, Advantages of FEC: SCALE, FEC Achieves These Advantages: SCANC, Hierarchy of Fog and Edge Computing, Business Models, Opportunities and Challenges.

**Addressing the Challenges in Federating Edge Resources:** Introduction, The Networking Challenge, The Management Challenge, Miscellaneous Challenges.

#### UNIT-II:

**Integrating IoT + Fog + Cloud Infrastructures:** System Modeling and Research Challenges: Introduction, Methodology, Integrated C2F2T Literature by Modeling Technique, Integrated C2F2T Literature by Use-Case Scenarios, Integrated C2F2T Literature by Metrics, Future Research Directions. Management and Orchestration of Network Slices in 5G, Fog, Edge, and Clouds: Introduction, Background, Network Slicing in 5G, Network Slicing in Software-Defined Clouds, Network Slicing, Management in Edge and Fog, Future Research Directions.

### **UNIT-III:**

**Optimization Problems in Fog and Edge Computing:** Introduction, Background / Related Work, Preliminaries, The Case for Optimization in Fog Computing, Formal Modeling Framework for Fog Computing, Metrics, Further Quality, Attributes, Optimization, Opportunities along the Fog Architecture, Optimization Opportunities along the Service Life Cycle, Toward a Taxonomy of Optimization Problems in Fog Computing, Optimization Techniques, Future Research Directions.

**Middleware:** Introduction, Need for Fog and Edge Computing Middleware, Design Goals, State-of-the-Art Middleware Infrastructures, System Model, Proposed Architecture, Case Study Example, Future Research Directions.

### **UNIT-IV:**

**Lightweight Container Middleware for Edge Cloud Architectures:** Introduction, Background/Related Work, Clusters for Lightweight Edge Clouds, Architecture Management – Storage and Orchestration, IoT Integration, Security Management for Edge Cloud Architectures, Future Research Directions.

**Data Management in Fog Computing:** Introduction, Background, Fog Data Management, Future Research and Direction.

### **UNIT-V:**

**Fog Computing Realization for Big Data Analytics:** Introduction, Big Data Analytics, Data Analytics in the Fog, Prototypes and Evaluation, Architecture, Configurations, Case Studies, Related Work, Future Research Direction.

Smart Surveillance Video Stream Processing at the Edge for Real-Time Human Objects

**Tracking:** Introduction, Human Object Detection, Object Tracking, Lightweight Human Detection, Case Study, Future Research Directions.

### **TEXT BOOK:**

1. Fog and Edge Computing: Principles and Paradigms (Series on Parallel and Distributed Computing), Rajkumar Buyya and Satish Narayana Srirama, Wiley

### **REFERENCES:**

1. Fog Computing: Theory and Practice, Assad Abbas, Samee U. Khan, Albert Y. Zomaya
2. Fog Computing: Helping the Internet of Things Realize its Potential, Amir Vahid Dastjerdi and Rajkumar Buyya, University of Melbourne